

8.1-3

Show that there is no comparison sort whose running time is linear for at least half of the $n!$ inputs of length n . What about a fraction of $1/n$ of the inputs of length n ? What about a fraction $1/2^n$?

8.3-4

Show how to sort n integers in the range 0 to $n^3 - 1$ in $O(n)$ time.

8.4-2

Explain why the worst-case running time for bucket sort is $\Theta(n^2)$. What simple change to the algorithm preserves its linear average-case running time and makes its worst-case running time $O(n \lg n)$?

9.3-3

Show how quicksort can be made to run in $O(n \lg n)$ time in the worst case, assuming that all elements are distinct.